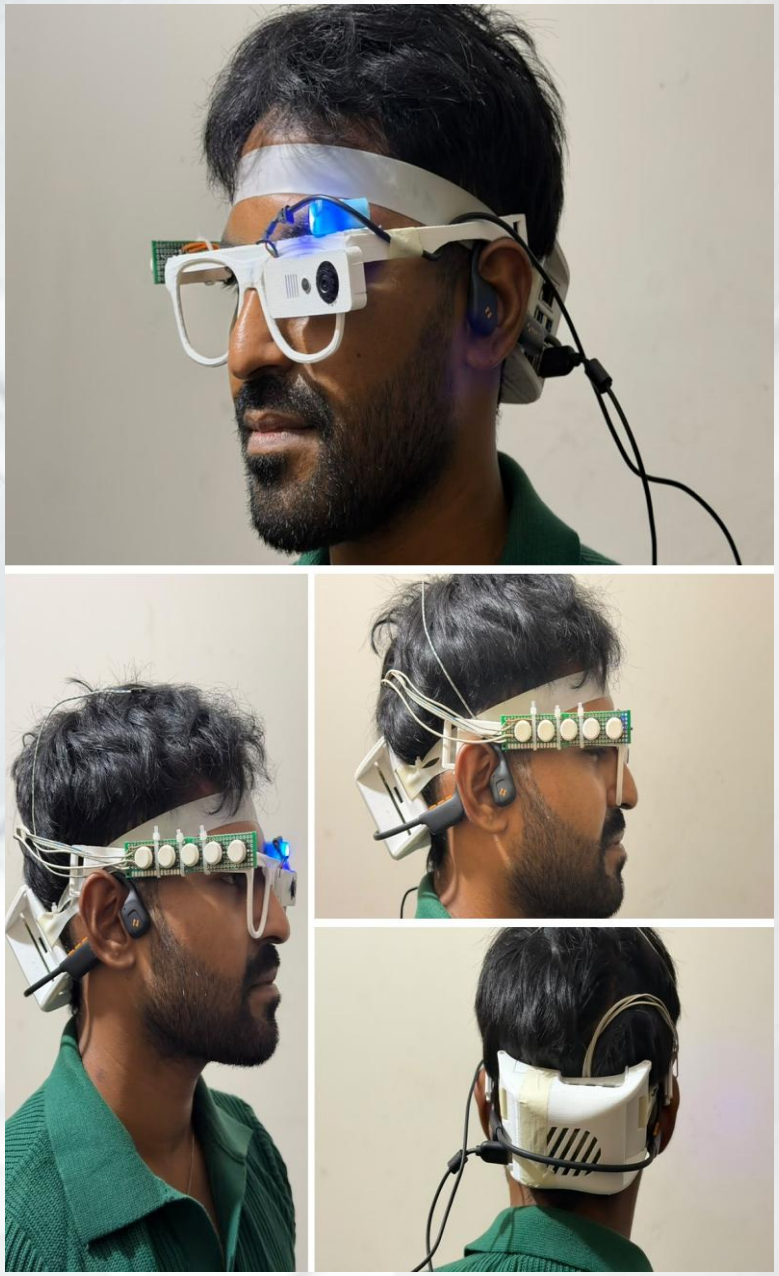
SARA · Smart Assistive Reading Aid

A Wearable Assistive Glass for Authenticating Bangladeshi Banknotes: Fusing Physical Security Features and Deep Learning for Visually Impaired Users

A bilingual (Bangla and English) smart glass that also reads printed Bangla and English text and detects everyday objects, offline on a Raspberry Pi 5

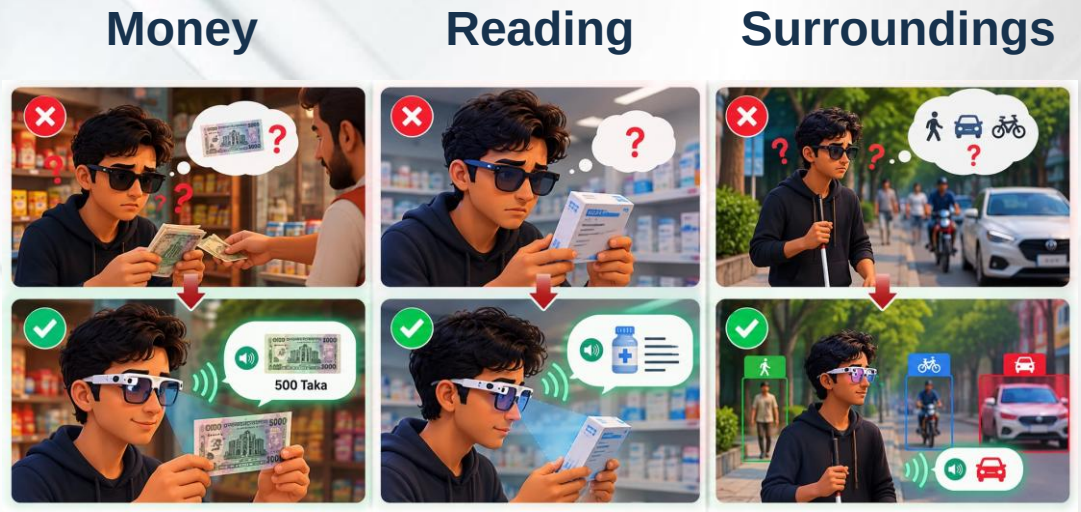
Supervisor:
Mohammad Shidujaman, PhD
Assistant Professor
Department of Computer Science & Engineering

Submitted By:
MD Abdun Fattah Lam (2220014)
Mukut Protim Memo (2020086)
Tanjila Hossain Tapti (2130561)



Introduction

- At least 2.2 billion people have a near or distance vision impairment, and its prevalence in low- and middle-income regions is about four times that in high-income regions [1]
- In Bangladesh, three everyday questions usually require help from another person:
  - **Money:** which note was handed back, and is it genuine?
  - **Reading:** what does a medicine label, sign or letter say?
  - **Surroundings:** what or who is ahead?
- In each case the blind person depends on an answer they cannot verify
- **SARA is an offline, bilingual assistive glass built for this setting**



Top row: today, an answer from another person. Bottom row: with SARA, the glass answers. (Illustration)

Problem Statement



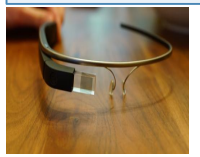
Existing aids do not close this gap

OrCam MyEye [2]



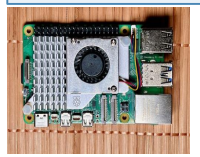
Offline, but no Bangla and no Taka

Research glasses [3, 4]



Pictures go to a server

Pi reading aids [5]



Mostly English-only

Counterfeit studies [6, 7, 8, 9, 10]



Not on a wearable

The gap

No reviewed device is offline, speaks Bangla and names Taka notes

Proposed Solution: SARA

User wearing SARA (staged)



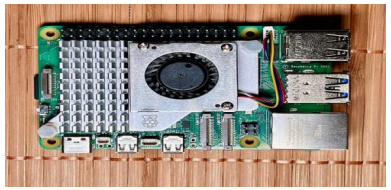
button, or "Glass, check money" [11]

Glass camera



captures a picture

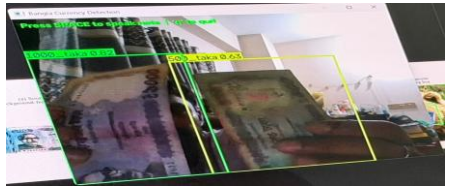
Raspberry Pi 5 [12]



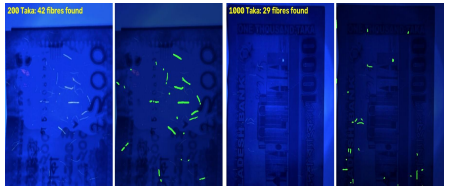
Offline: text, objects, Taka, emotion
Online (optional): scene description only

processes offline

Taka recognition [13]



Authenticity: watermark + UV



Text reading [14, 15]



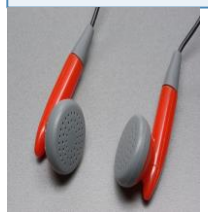
Objects and people [13, 16]



Safe verdict (money)

"Likely genuine" or "check by hand whether it is genuine"; it never says "counterfeit"

Spoken answer in Bangla or English [17] + a short vibration

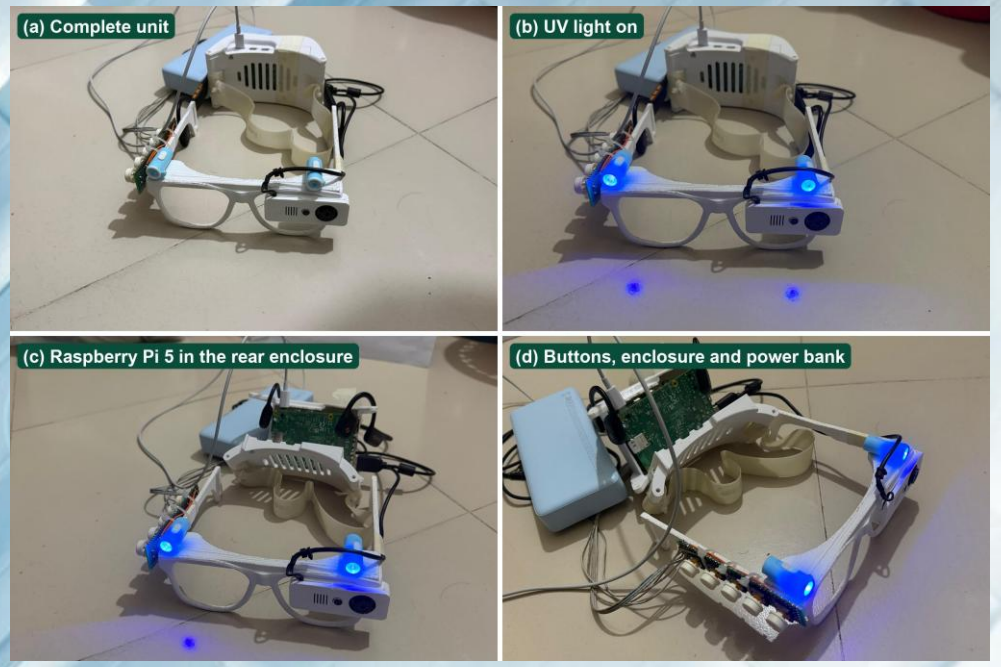




পাঁচশত টাকার নোট

"500 Taka note": known without asking anyone

System Features



Prototype version 2: the complete unit, the UV light on, the Pi 5 in the rear enclosure, buttons and power bank

Text reading

Bangla and English

Objects

Named in Bangla

Taka notes

Value + safe check

Emotion

Face in front

Online description

Optional; needs internet

How the user starts a mode: five buttons on the frame, or a voice command: "Glass, read text"

Models: text [14, 15] CRAFT + EasyOCR; objects [13, 16] YOLOv8 on COCO; emotion [18] RAF-DB; voice commands [11] Vosk.

Research Questions

1 System

Can it all run on a Raspberry Pi 5?

2 Synthetic data

Can unlabelled training data detect real notes?

3 Number of views

Can one view of a note be enough?

4 Unseen prints

Does the check hold on new counterfeit prints?

5 Safety

Can it never accuse a genuine note?

6 Bangla OCR

What matters most for reading Bangla?

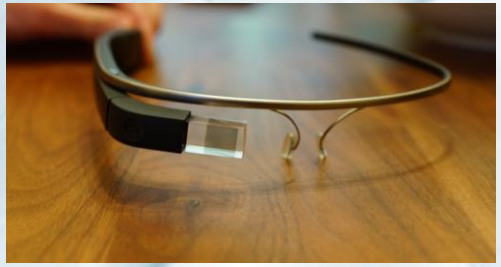
Literature Review: Existing Assistive Devices

System	Reads Bangla	Knows Taka	Works offline	Wearable
OrCam MyEye [2]	No	No	Yes	Yes
Google Glass scene analysis [3]	No	No	No	Yes
AIris (2024) [4]	No	No	No	Yes
Raspberry Pi reading aids *	No	No	Partly	Partly
SARA (ours)	**Yes**	**Yes**	**Yes**	**Yes**

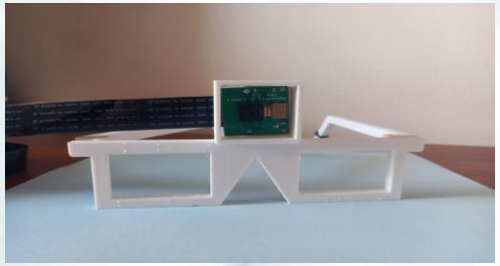
*Rows with a reference: from the cited source. * Our assessment of typical hobby builds, not a cited result.*



OrCam MyEye [2]



Google Glass analysis [3]



AIris [4]

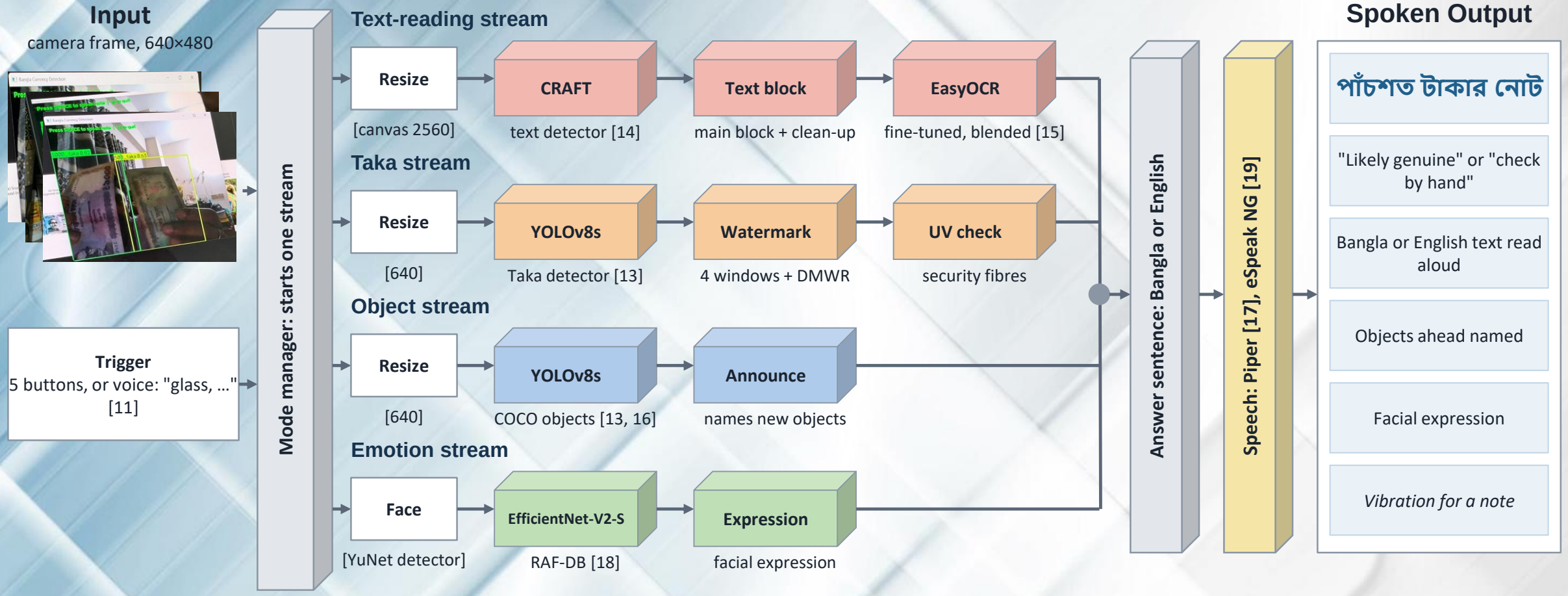
The research gap

None of the systems we reviewed combines Bangla, Taka and offline use.

- Counterfeit tests reuse the same prints
- Few report speed on the real device

Photos: OrCam MyEye by Ylip, CC BY-SA 4.0; Google Glass by Ted Eytan, CC BY-SA 2.0 (Wikimedia Commons); AIris from [4]. Comparison from [2, 3, 4].

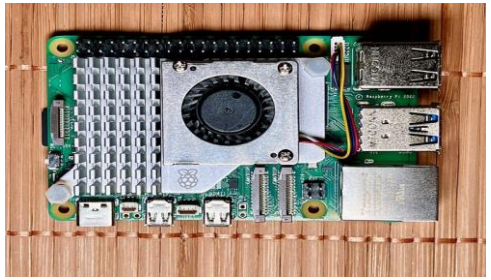
System Architecture



All streams run on the Raspberry Pi 5 CPU, offline. The optional online scene-description mode is not shown.

Sources: [11] Vosk; [13] Ultralytics YOLOv8; [14] Baek et al. 2019; [15] EasyOCR; [16] COCO; [17] Piper; [18] RAF-DB; [19] eSpeak NG.

Hardware Design



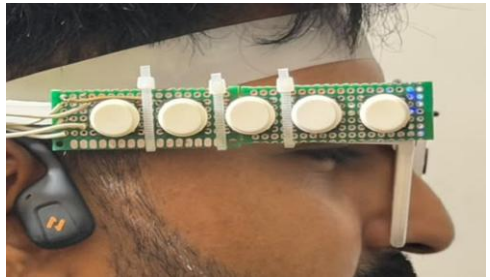
Raspberry Pi 5

4-core 2.4 GHz, 8 GB; runs every model [12]



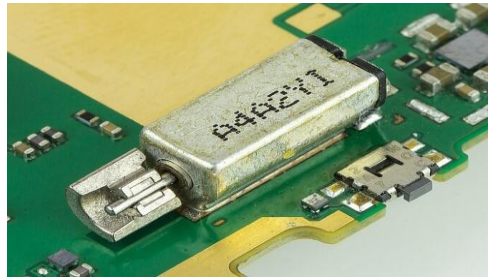
Camera Module 2

8 MP IMX219, used at 640×480



Five push-buttons

MODE, ACTION, READ, VOL+, VOL−



Vibration motor

A buzz when a note is found



Bone-conduction headphone

Speech; the ears stay free



Power bank

USB-C supply for the Pi 5



Frame and UV lamp

3D-printed; UV-A 360–395 nm

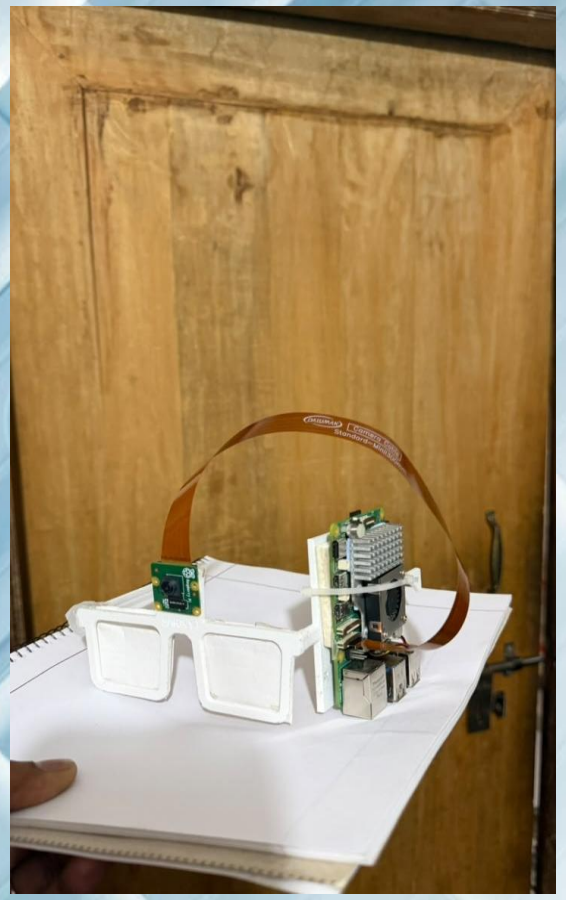


Rear enclosure

Ventilated case for the Pi 5

Pictures (Wikimedia Commons): Raspberry Pi 5 by SimonWaldherr, CC BY 4.0; Camera Module v2 by the Raspberry Pi Foundation, CC BY-SA 4.0; vibration motor by Raimond Spekking, CC BY-SA 4.0; power bank by Pinguinguy, CC0. Buttons, headphone, frame and enclosure: our prototype.

Evolution of the Prototype



Version 1
Board on one temple



Front
Camera + UV lamp



Side
5 buttons, bone-conduction earphone



Rear
Pi 5 in a ventilated case

Banknote Detection: Synthetic Training Data



Note photos [20] are pasted onto real scenes [16], and the program writes every label itself.

SYNTHETIC COMPOSITES

0.995

mAP@0.5 on 2,282 held-out composites

REAL PHOTOS

94.1%

right note value on 1,536 photos taken by other people (laptop)

Sources: [20] BanglaTaka (Mendeley); [16] Lin et al., COCO, ECCV 2014; detector [13] Ultralytics YOLOv8. 94.1% with the hand-held detector (next slide).

Banknote Detection: Hand-Held Notes

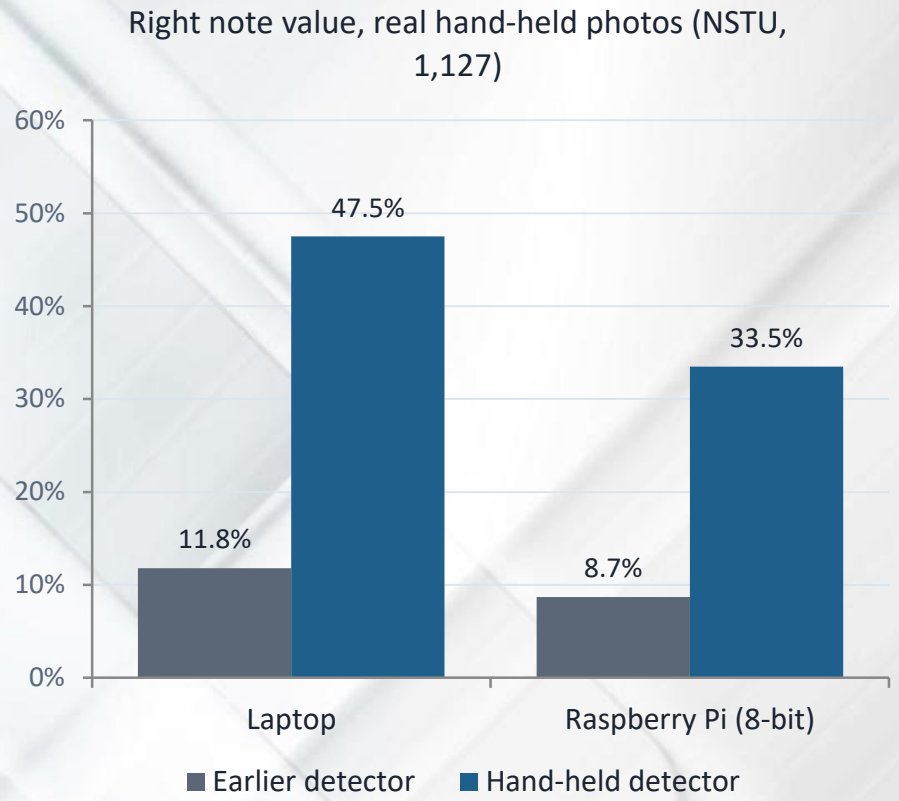
TRAINING: 6,000 SYNTHETIC CLOSE-UPS



TEST: REAL HAND-HELD PHOTOS

11.8% → 47.5%

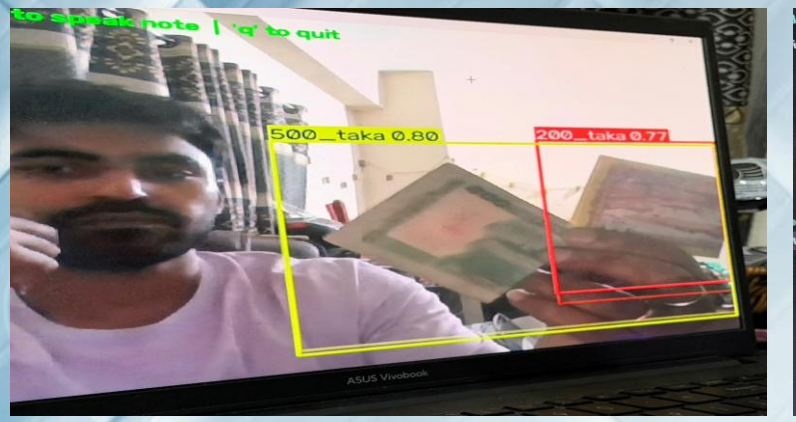
Right note value on the laptop. Trade-off: with 3 notes in view, the total is right 97.0% (was 99.2%).

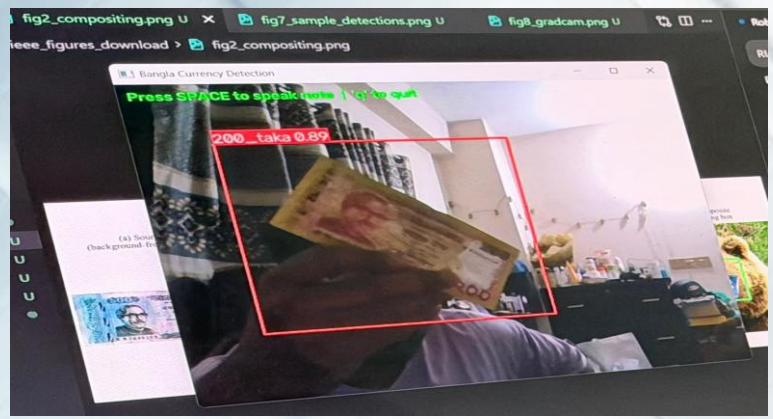


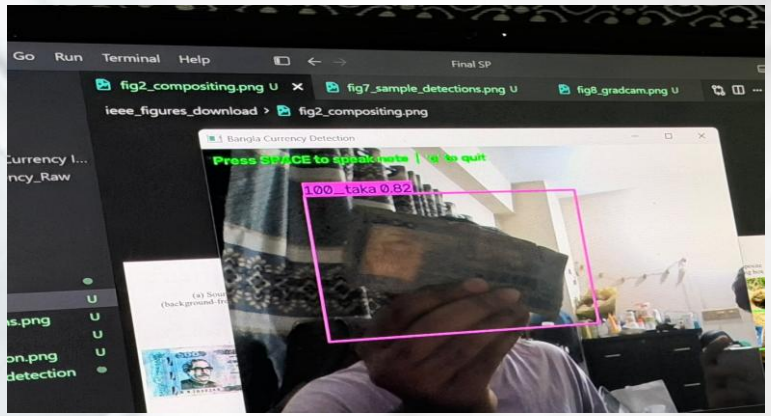
Sources: test photos [21] Rafi et al., NSTU-BDTAKA, Data in Brief 2024; detector [13] YOLOv8. Our results: paper_evidence/HANDHELD_FINETUNE.md.

Live Detection Tests

Informal demonstrations, not a measured accuracy study







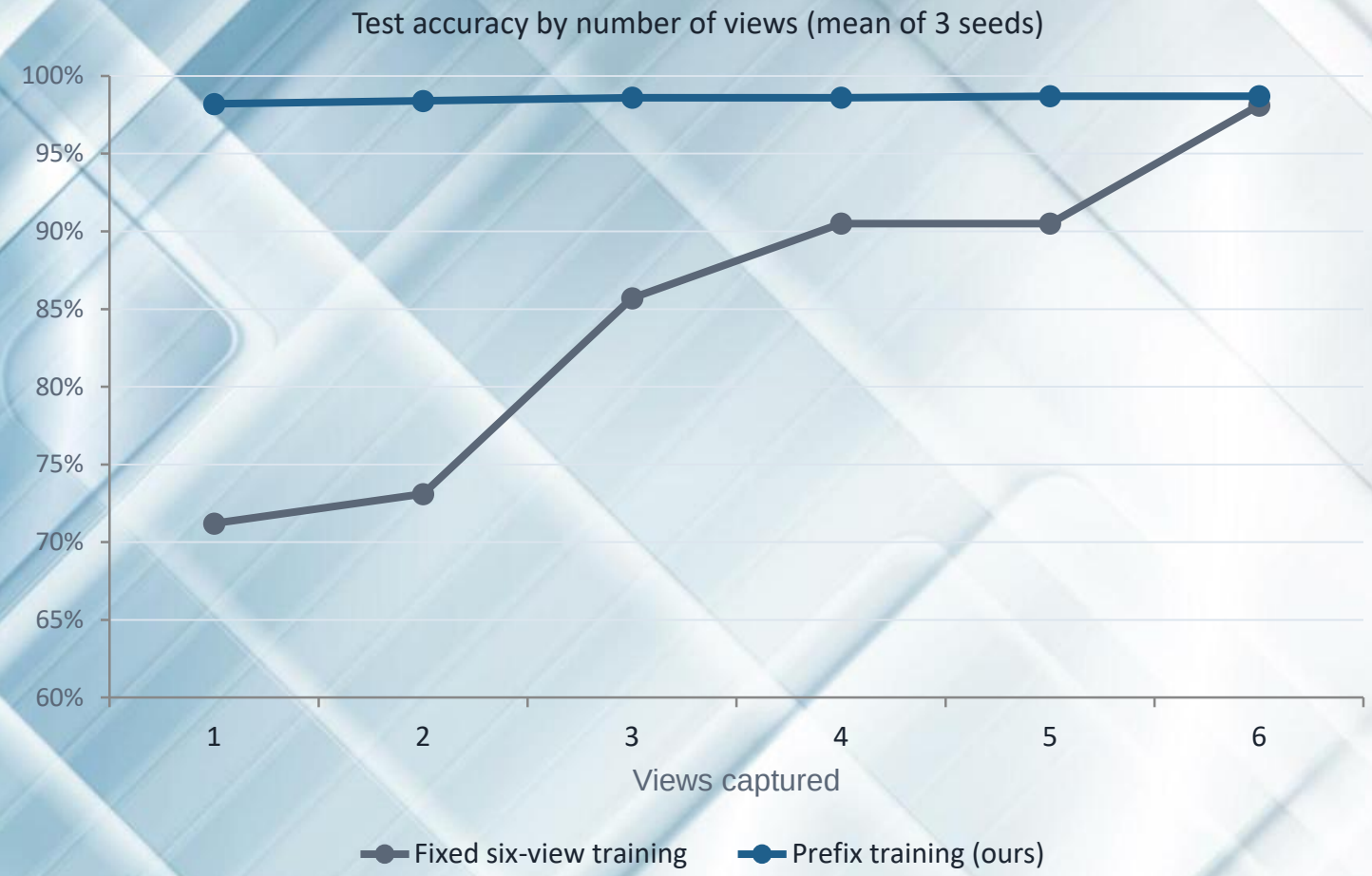






Six note values (10 to 1,000 Taka) named at 0.63–0.89 confidence, including worn and folded notes. Laptop webcam and the first glass prototype.

Counterfeit Screening: Prefix-Robust Multi-View Training



With a single view

71.2%

rises to

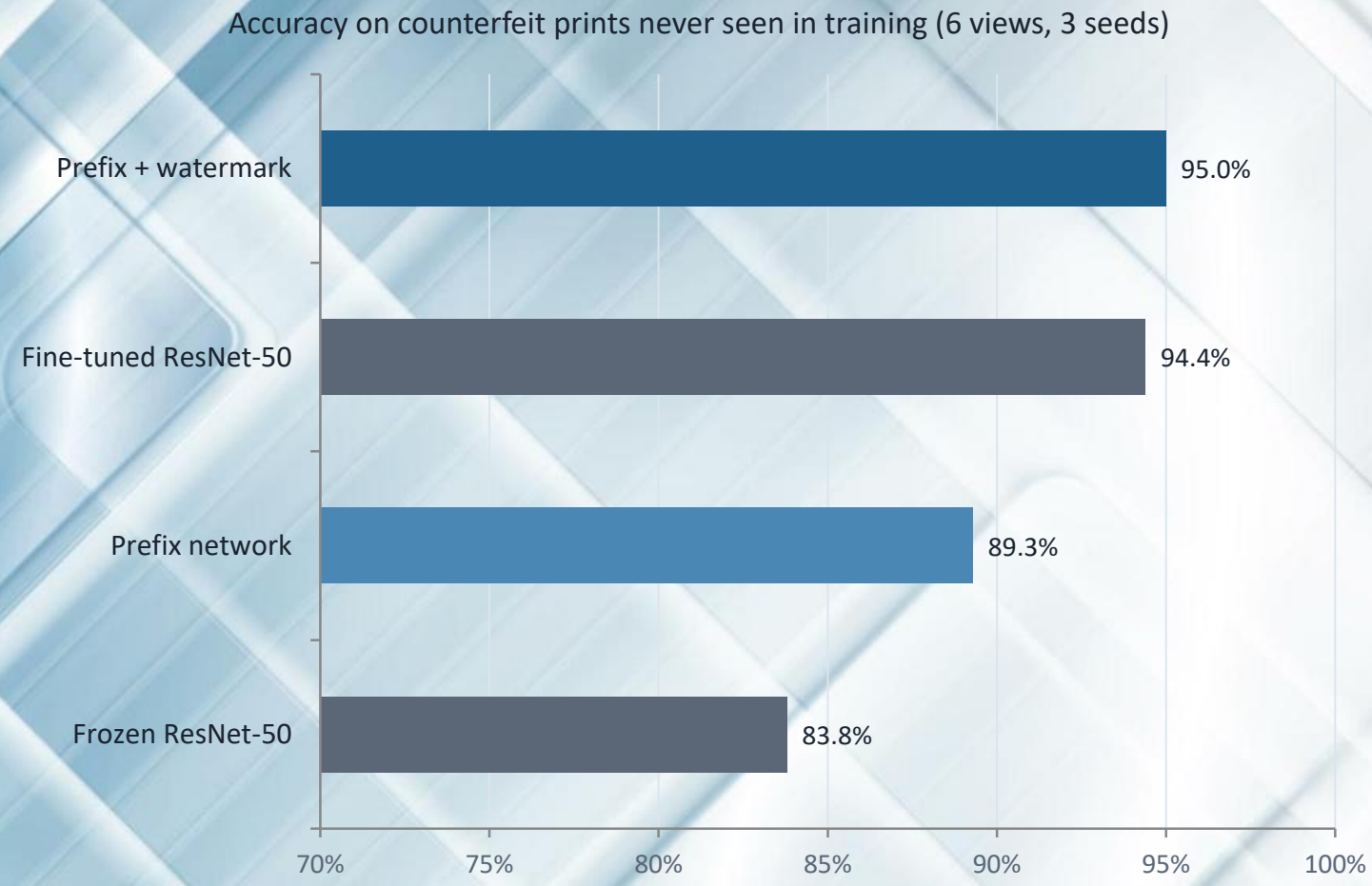
98.2%

Training on random first views: one view is enough.

Held-out JaalTaka test notes, never used for training or for any choice

Data: [9, 10] Piyas et al., JaalTaka dataset and article, 2025-2026 (1,390 notes, six views each; counterfeits seized by the Rapid Action Battalion).

Counterfeit Screening: Evaluation on Unseen Prints



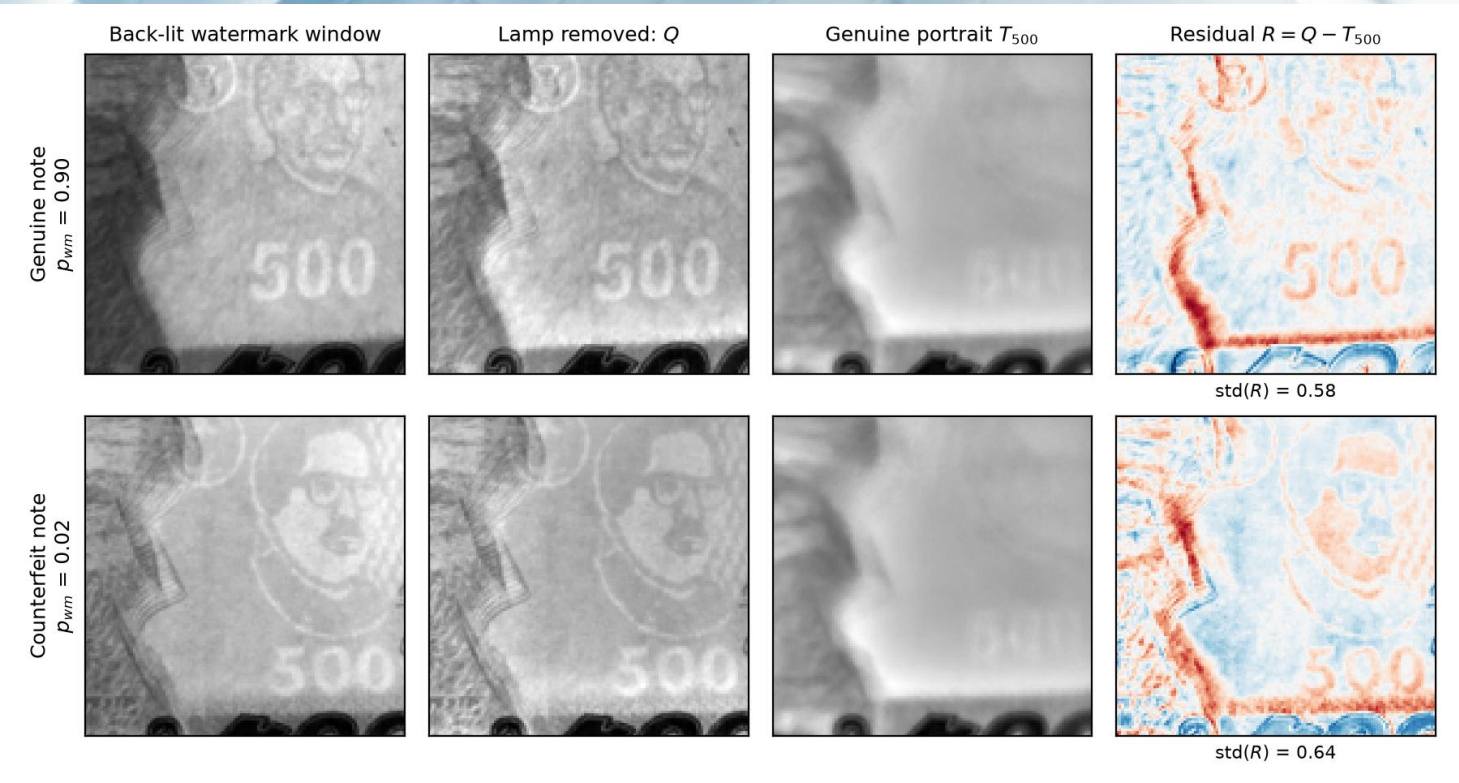
95.0%

on counterfeit prints never seen in training, with the watermark check

Why a stricter split: 279 of 322 readable 500 Taka counterfeits share one serial number, so the usual split tests copies of known prints.

Data: [9] JaalTaka; serial-number audit and print-disjoint split are ours. Earlier Taka counterfeit work: [6] SVM on security features; [7] CNNs; [8] UV flecks + RPN.

Watermark Verification: DMWR



1. Remove the lamp: Q = I / (G∗I)
2. Genuine portrait: T = median of Q over genuine notes
3. Residual R = Q − T → two small CNN streams → "watermark clear?"

91.5%

on unseen counterfeit prints (3 seeds, ±0.3)

46× smaller

than MobileNetV2 [22] (47,818 parameters, 0.21 MB), and as accurate: 92.9%, p ≥ 0.375

Baseline [22] Sandler et al., MobileNetV2, CVPR 2018; data [9] JaalTaka. DMWR (denomination-matched watermark residual) is our method.

Safe Verdict Policy



0 / 88

test counterfeits passed as "likely genuine"

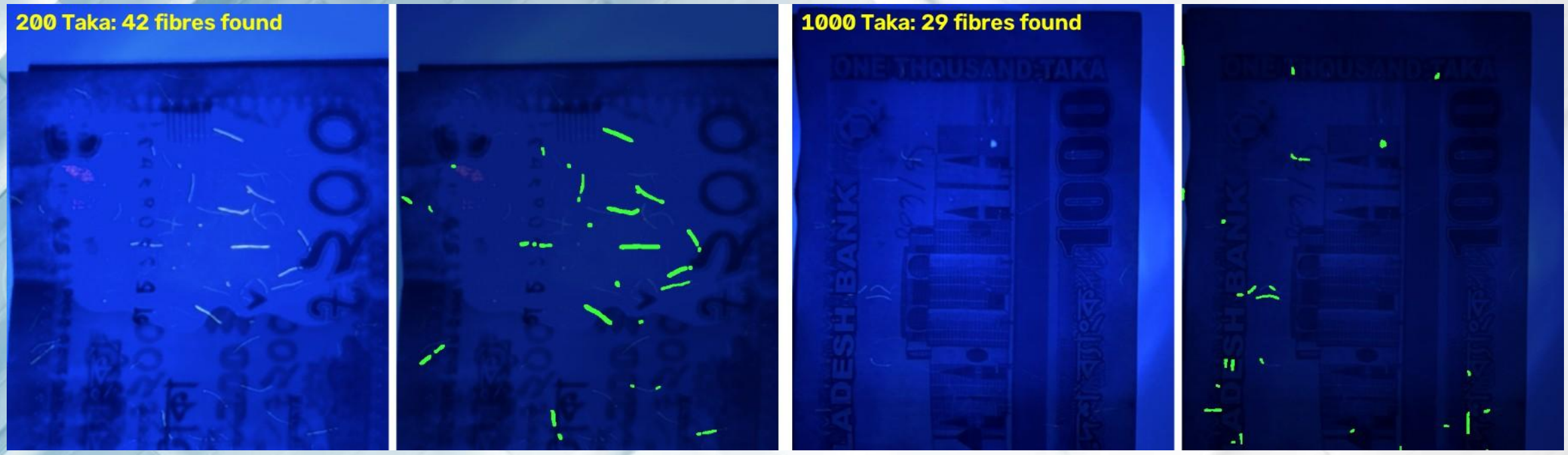
"Likely genuine": the score is higher than every counterfeit seen in validation (57.5% of genuine notes).

"Check by hand": the glass is not sure. It never means "counterfeit".

A screening aid, not a guaranteed counterfeit detector

Test notes: [9] JaalTaka.

UV Security-Fibre Verification



85.6%

of 111 real UV photos of genuine notes accepted

88.6%

of SIMULATED fibre-free sheets rejected (not real counterfeits)

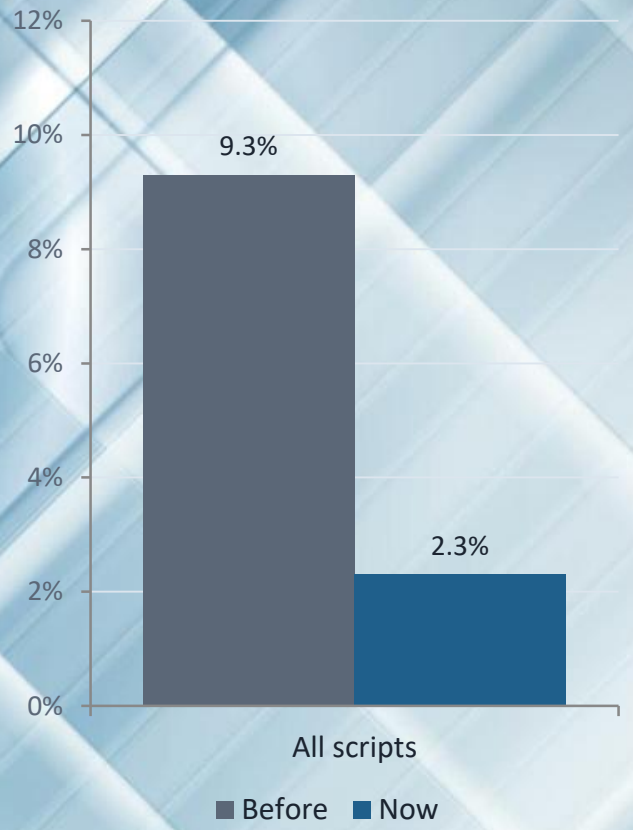
Prototype, off by default

The rejection result is simulated; it must be validated on real counterfeit notes

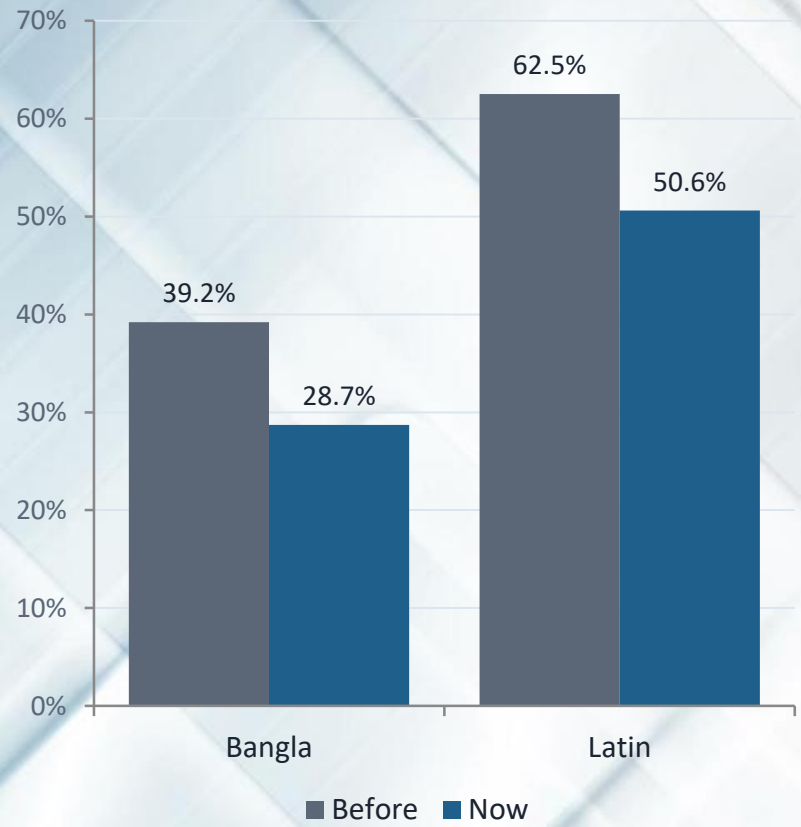
The glass's lamp emits 360-395 nm, covering the 365 nm used for banknote checks; banknote paper stays dark while its fibres glow [23]. Prior Taka UV work: [8] (96.3%). Fibre check and photos are ours.

Bangla and English Text Recognition

Rendered text (clean)



Real word photos (hard)



What helped

- Reading the main text block only
- Tuned detector and clean-up
- Recogniser fine-tuned on real words

What did not

Binarisation, Tesseract, TrOCR, PaddleOCR, dictionaries

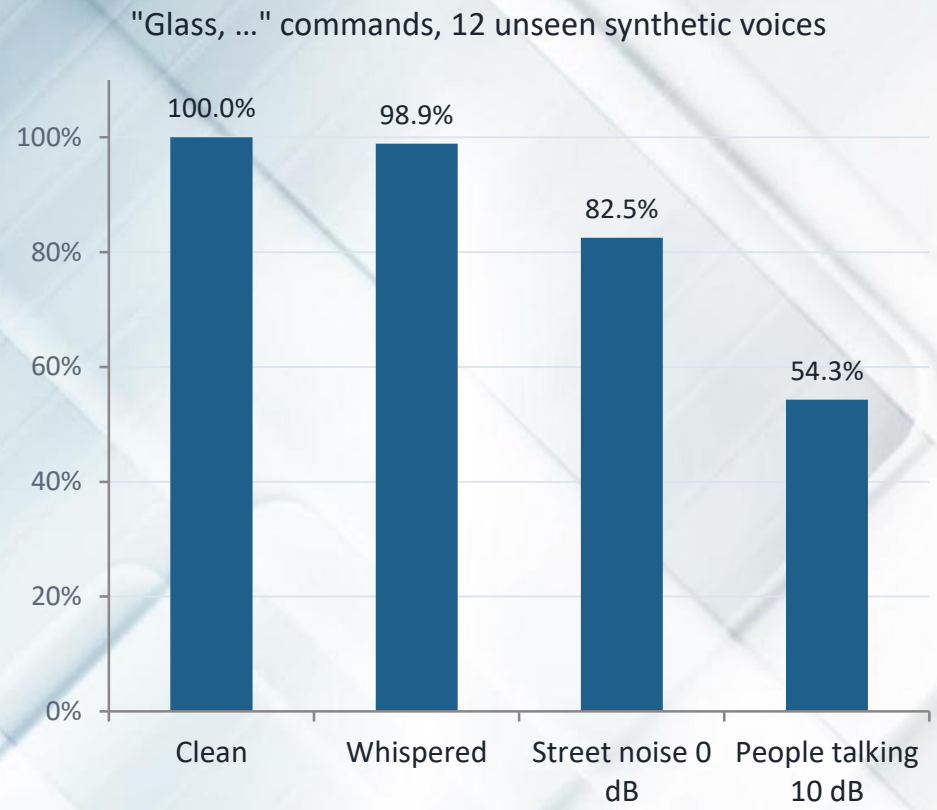
Character error rate, lower is better. Measured on evaluation datasets, not yet through the glass.

Engine [15] EasyOCR with [14] CRAFT; real words from [24] ICDAR 2019 MLT (1,000 test words). Every choice made on validation data.

Voice Interaction

Speaking

- Natural Piper voice, Bangla and English [17, 19]
- No broken sound when the Pi is busy
- "পড়ছি" (reading) while text is captured



212 → 13

false commands in 5,040 ordinary sentences, thanks to the wake word

0.86 s

from the end of speech to the action (median)

Command results: synthetic voices on a laptop, not human trials on the glasses

Recogniser [11] Vosk; noise from [25] Piczak, ESC-50; voices [17, 19]. Results: paper_evidence/VOICE_COMMANDS_FAST.md.

Performance on the Raspberry Pi 5

399 ms
currency check, end to end

64 ms
watermark check

562 ms
object scan (YOLOv8s)

680 ms
emotion, face in view

18.3 s
3-line sign, text found once

1.48 GB
peak memory, 30-minute run



Key finding: heat

88.7 °C

peak temperature; the Pi throttled in 76% of samples (idle 71 °C)

Response: a thermal governor slows background scans when hot; better cooling is future work.

Measured on a Raspberry Pi 5 (8 GB) on 13 October 2026, board at its temperature limit; memory from the 1 October run. Changes after 13 October are not yet timed on the Pi.

Board: [12] Raspberry Pi 5 (8 GB). All timings are ours (benchmark_report/, 1 and 13 October 2026).

Prototype Testing and Project Status



Prototype version 2 in a staged shop exchange by team members; illustration only, no error rate counted.

DONE — **Engineering checks**

Each part tested on held-out data, on a laptop and the Pi 5

PRELIMINARY — **First field test**

Glass camera: currency unreliable, Bangla text not read

IN PROGRESS — **Participant trials**

Sessions with a visually impaired participant have started

NONE YET — **Full user study**

No results about users are claimed

Future Directions



Camera

Higher resolution or autofocus

User study

8–12 participants, including their own voice commands

More banknotes

Unseen counterfeit prints; UV check on real fakes

Speed and cooling

Time the new code on the Pi; better cooling; battery

Done: engineering tests **In progress:** participant trials **Future:** the four steps above

Conclusion



Built

An offline Raspberry Pi 5 glass that reads Bangla and English, names objects and Taka notes, and speaks Bangla

Investigated

Synthetic training data, prefix training, unseen prints, a safe verdict, DMWR, hand-held notes, OCR, voice

Still to validate

Real counterfeits (also under UV), people's voices, the glass camera, a completed user study

THANK YOU

Questions?

References

[1] World Health Organization. Blindness and vision impairment: fact sheet, updated 10 February 2026.

[2] OrCam Technologies. OrCam MyEye.

[3] Ali A., H. et al. A Google Glass based real-time scene analysis for the visually impaired. IEEE Access 9, 2021.

[4] Brilli, D. D. et al. AIris: an AI-powered wearable assistive device for the visually impaired. arXiv:2405.07606, 2024.

[5] Adusumilli, V. et al. Reading aid and translator with Raspberry Pi for blind people. ICACCS, 2023, pp. 327-331.

[6] Uddin, M. S., Das, P. P., Roney, M. S. A. Image-based approach for the detection of counterfeit banknotes of Bangladesh. ICIEV, 2016.

[7] Pathak, A. et al. Enhanced counterfeit detection of Bangladesh currency through convolutional neural networks. IJIRCST 12(2), 2024.

[8] Chowdhury, A. A. et al. Evaluation of deep learning models on UV ink: a fake money detection scheme with RPN. ICCA (ACM), 2022.

[9] Piyas, B. R. C. et al. JaalTaka: a benchmark dataset for detecting counterfeit Bangladeshi banknotes. Mendeley Data V2, 2025.

[10] Piyas, B. R. C. et al. JaalTaka: a benchmark dataset for detecting counterfeit Bangladeshi banknotes. Data in Brief 64, 112377, 2026.

[11] Alpha Cephei. Vosk offline speech recognition toolkit.

[12] Raspberry Pi Ltd. Raspberry Pi 5 product brief, 2023.

[13] Jocher, G., Qiu, J., Chaurasia, A. Ultralytics YOLO, version 8.0.0, 2023.

[14] Baek, Y. et al. Character region awareness for text detection (CRAFT). CVPR, 2019.

[15] JaidedAI. EasyOCR: ready-to-use OCR with 80+ languages.

[16] Lin, T.-Y. et al. Microsoft COCO: common objects in context. ECCV, 2014.

[17] Rhasspy. Piper: a fast, local neural text-to-speech system.

[18] Li, S., Deng, W., Du, J. Reliable crowdsourcing and deep locality-preserving learning for expression recognition in the wild (RAF-DB). CVPR, 2017.

[19] eSpeak NG text-to-speech.

[20] Nuhash, M. N. I., Akter, S., Mojumdar, M. U. A diverse image dataset for Bangladeshi currency recognition (BanglaTaka). Mendeley Data V1, 2024.

[21] Rafi, M. J. A., Rony, M., Majadi, N. NSTU-BDTAKA: an open dataset for Bangladeshi paper currency detection and recognition. Data in Brief 55, 2024.

[22] Sandler, M. et al. MobileNetV2: inverted residuals and linear bottlenecks. CVPR, 2018.

[23] Mutanen, J., Jaaskelainen, T., Parkkinen, J. P. S. Luminescent security properties of banknotes. IS&T PICS Conference, 2003, pp. 421-424.

[24] Nayef, N. et al. ICDAR2019 robust reading challenge on multi-lingual scene text (RRC-MLT-2019). ICDAR, 2019.

[25] Piczak, K. J. ESC: dataset for environmental sound classification. ACM Multimedia, 2015.

Resources and Links

Our code and results

SARA: glass software for the Raspberry Pi 5
github.com/aflam330/SARA

Research repository: experiments, results, thesis
github.com/aflam330/Senior_project_wearable_glass_for_blind_people_using_ocr_and_ylo

Datasets

JaalTaka (counterfeit notes)
data.mendeley.com/datasets/2m7wk5cy4c/2

BanglaTaka (note photos)
data.mendeley.com/datasets/3cv2sypkkh/1

NSTU-BDTAKA (hand-held notes)
www.sciencedirect.com/science/article/pii/S2352340924006681

Microsoft COCO
cocodataset.org

ICDAR 2019 MLT
rrc.cvc.uab.es/?ch=15

ESC-50 environmental sounds
github.com/karolpiczak/ESC-50

Open-source tools

Ultralytics YOLOv8
github.com/ultralytics/ultralytics

EasyOCR
github.com/JaidedAI/EasyOCR

Piper text-to-speech
github.com/rhasspy/piper

Vosk speech recognition
alphacephei.com/vosk/

eSpeak NG
github.com/espeak-ng/espeak-ng

WHO vision fact sheet
www.who.int/news-room/fact-sheets/detail/blindness-and-visual-impairment